Supplementary Material for “Exact Structural Abstraction and Tractability Limits”

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
FR1	`currentFamilies_partition_by_role` `Paper4dFrontier/Classification.lean`	FR2	`currentFamilies_partition_counts` `Paper4dFrontier/Classification.lean`
FR3	`current_families_have_finite_basis` `Paper4dFrontier/Classification.lean`	FR4	`current_positive_interfaces_factor_through_role` `Paper4dFrontier/Classification.lean`
FR5	`lifted_families_reduce_to_core` `Paper4dFrontier/Classification.lean`	FR6	`degenerateFamilies_complete` `Paper4dFrontier/Classification.lean`
FR7	`exists_decisionProblem_realizing_labeling` `Paper4dFrontier/Realizability.lean`	FR8	`realizingProblem_decisionEquiv_iff` `Paper4dFrontier/Realizability.lean`
FR13	`isOptimal_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR14	`opt_eq_positiveAffineTransform` `Paper4dFrontier/FamilyAxioms.lean`
FR15	`isSufficient_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR16	`isRelevant_positiveAffineTransform_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR17	`decisionEquiv_relabelActions_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR18	`decisionEquiv_relabelStates_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR19	`isSufficient_relabelActions_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR20	`isSufficient_relabelStates_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR21	`allProblems_relabelInvariant` `Paper4dFrontier/FamilyAxioms.lean`	FR22	`allProblems_kernelUniversal` `Paper4dFrontier/FamilyAxioms.lean`
FR23	`relabelInvariant_not_enough` `Paper4dFrontier/FamilyAxioms.lean`	FR24	`optRangeCard_realizingIdentity` `Paper4dFrontier/FamilyAxioms.lean`
FR25	`currentFamilies_explicit` `Paper4dFrontier/Classification.lean`	FR26	`coreFamilies_explicit` `Paper4dFrontier/Classification.lean`
FR27	`liftedFamilies_explicit` `Paper4dFrontier/Classification.lean`	FR28	`degenerateFamilies_explicit` `Paper4dFrontier/Classification.lean`
FR29	`primitiveMechanisms_explicit` `Paper4dFrontier/Classification.lean`	FR30	`isSufficient_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR31	`isRelevant_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR32	`isIrrelevant_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR33	`isMinimalSufficient_iff_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR34	`sufficientSets_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR35	`relevantSet_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR37	`quotientCard_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR38	`quotientCard_realizingIdentity` `Paper4dFrontier/FamilyAxioms.lean`	FR39	`realizingProblemQuotientEquivLabelRange_apply_quotientMap` `Paper4dFrontier/Realizability.lean`

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR40	`realizingProblem_quotientCard_eq_labelRangeCard` `Paper4dFrontier/Realizability.lean`	FR41	`certificationStatistic_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR42	`sufficientSetCount_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR43	`minimalSufficientSetCount_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`
FR44	`relevantCoordCount_eq_of_decisionEquiv_iff` `Paper4dFrontier/FamilyAxioms.lean`	FR45	`setoidRealizingProblem_decisionEquiv_iff` `Paper4dFrontier/Realizability.lean`
FR46	`setoidRealizingProblemQuotientEquivSetoidQuotient_apply_quotientMap` `Paper4dFrontier/Realizability.lean`	FR47	`exists_decisionProblem_realizing_setoid` `Paper4dFrontier/Realizability.lean`
FR48	`isSufficient_iff_agreementRel_le_decisionEquiv` `Paper4dFrontier/FamilyAxioms.lean`	FR49	`isIrrelevant_iff_sufficient_erase` `Paper4dFrontier/FamilyAxioms.lean`
FR50	`isRelevant_iff_not_sufficient_erase` `Paper4dFrontier/FamilyAxioms.lean`	FR51	`quotientEquiv_of_decisionEquiv_iff_apply_quotientMap` `Paper4dFrontier/FamilyAxioms.lean`
FR52	`empty_sufficient_of_constant_opt` `Paper4dFrontier/ClosureLaws.lean`	FR53	`irrelevant_of_constant_opt` `Paper4dFrontier/ClosureLaws.lean`
FR54	`decisionEquiv_duplicateAction_iff` `Paper4dFrontier/ClosureLaws.lean`	FR55	`isSufficient_duplicateAction_iff` `Paper4dFrontier/ClosureLaws.lean`
FR56	`isRelevant_duplicateAction_iff` `Paper4dFrontier/ClosureLaws.lean`	FR57	`decisionEquiv_duplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`
FR58	`isSufficient_duplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`	FR59	`isRelevant_duplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`
FR60	`duplicateStateQuotientEquivOriginal_apply_quotientMap` `Paper4dFrontier/ClosureLaws.lean`	FR66	`booleanCube_card` `Paper4dFrontier/EnumerationBounds.lean`
FR67	`exists_booleanCube_larger_than_monomial` `Paper4dFrontier/EnumerationBounds.lean`	FR68	`decisionEquiv_addDuplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`
FR69	`isSufficient_addDuplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`	FR70	`isRelevant_addDuplicateState_iff` `Paper4dFrontier/ClosureLaws.lean`
FR71	`addDuplicateStateQuotientEquivOriginal_apply_quotientMap` `Paper4dFrontier/ClosureLaws.lean`	FR72	`some_mem_opt_addDuplicateAction_iff` `Paper4dFrontier/ClosureLaws.lean`
FR73	`none_mem_opt_addDuplicateAction_iff` `Paper4dFrontier/ClosureLaws.lean`	FR83	`isSufficient_noiseExtended_iff` `Paper4dFrontier/DimensionalNoiseExtension.lean`
FR84	`isRelevant_noiseExtended_iff` `Paper4dFrontier/DimensionalNoiseExtension.lean`	FR85	`lastCoord_irrelevant_noiseExtended` `Paper4dFrontier/DimensionalNoiseExtension.lean`
FR100	`parentTree_realTreewidth_one` `Paper4dFrontier/ParentTreewidth.lean`	FR101	`low_rank_only_witness` `Paper4dFrontier/LowRankOnlyWitness.lean`
FR105	`separable_only_witness` `Paper4dFrontier/SeparableOnlyWitness.lean`	FR106	`bounded_actions_only_witness` `Paper4dFrontier/BoundedActionsOnlyWitness2.lean`
FR108	`all_independent_core_mechanisms_nondegenerate` `Paper4dFrontier/Block2Progress.lean`	FR110	`parentTreeStructured_implies_treeStructured` `Paper4dFrontier/ParentTreewidth.lean`
FR111	`treeStructured_and_unique_parent_iff_parentTreeStructured` `Paper4dFrontier/ParentTreewidth.lean`	FR112	`bounded_treewidth_only_witness` `Paper4dFrontier/BoundedTreewidthOnlyWitness.lean`
FR113	`symmetry_only_witness` `Paper4dFrontier/SymmetryOnlyWitness.lean`	FR114	`weak_tree_structured_not_width_one` `Paper4dFrontier/LegacyTreeGap.lean`
FR115	`tree_structure_hardness_bridge` `Paper4dFrontier/TreeStructureHardnessBridge.lean`	FR118	`pairCrossDifference_eq_binaryCrossDifference_of_lt` `Paper4dFrontier/BinaryPairwiseDichotomy.lean`
FR119	`pairwise_zero_crossDifference_unaryDecomposition` `Paper4dFrontier/BinaryPairwiseDichotomy.lean`	FR120	`binary_pairwise_symmetry_dichotomy` `Paper4dFrontier/BinaryPairwiseDichotomy.lean`
FR121	`actionGapCrossDifference_eq_binaryCrossDifference_of_lt` `Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean`	FR122	`decisionRelevant_zero_actionGap_implies_unaryReduction` `Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean`
FR123	`binary_pairwise_symmetry_decision_relevant_dichotomy` `Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean`	FR124	`block6_genuine_interaction_dichotomy_obstruction` `Paper4dFrontier/Block6Obstruction.lean`

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR125	decisionRelevantInteractionGraph_ addActionOffset Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR126	action_offset_can_force_constant_optimizer Paper4dFrontier/Block6Obstruction.lean
FR127	block6_action_offset_obstruction Paper4dFrontier/Block6Obstruction.lean	FR128	offsetNormalizedDecisionRelevantInteractionGraph_ wellDefined Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean
FR129	block6_offset_normalized_obstruction Paper4dFrontier/Block6Obstruction.lean	FR130	neverOptimalGhost_decisionRelevantGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR131	neverOptimalGhost_supportedGraph_eq_bot Paper4dFrontier/Block6Obstruction.lean	FR132	support_filtering_removes_known_block6_ obstructions Paper4dFrontier/Block6Obstruction.lean
FR133	block6_ghost_action_obstruction Paper4dFrontier/Block6Obstruction.lean	FR134	marginMasking_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean
FR135	marginMasking_zero_sufficient Paper4dFrontier/Block6Obstruction.lean	FR136	block6_optimizer_supported_obstruction Paper4dFrontier/Block6Obstruction.lean
FR137	MarginBounded Paper4dFrontier/DecisionRelevantPairwiseDichotomy.lean	FR138	dominantPair_marginBounded Paper4dFrontier/Block6Obstruction.lean
FR139	dominantPair_supportedGraph_eq_top Paper4dFrontier/Block6Obstruction.lean	FR140	block6_margin_bounded_obstruction Paper4dFrontier/Block6Obstruction.lean
FR145	not_collapseLandscapeInfinity_of_oracle_ predicate Paper4dFrontier/MetaCharacterization.lean	FR176	DecisionQuotient.DecisionProblem.quotient_is_ coarsest paper4/DecisionQuotient/Quotient.lean
FR177	DecisionQuotient.DecisionProblem.quotient_has_ unique_factorization paper4/DecisionQuotient/Quotient.lean	FR178	DecisionQuotient.DecisionProblem.srank_eq_ relevant_card paper4/DecisionQuotient/Tractability/ StructuralRank.lean
FR179	DecisionQuotient.quotientEntropy_le_srank_ binary paper4/DecisionQuotient/Information.lean	FR180	DecisionQuotient.Statistics.fisherMatrix_rank_ eq_srank paper4/DecisionQuotient/Statistics/ FisherInformation.lean
FR181	DecisionQuotient.Information.compression_ below_srank_fails paper4/DecisionQuotient/Information/ RDSrank.lean	FR182	DecisionQuotient.Information.srank_bits_ sufficient paper4/DecisionQuotient/Information/ RDSrank.lean
FR183	DecisionQuotient.Physics.single_future_zero_ cost paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean	FR184	DecisionQuotient.Physics.transportCost_pos_of_ offDiag paper4/DecisionQuotient/Physics/ WassersteinIntegrity.lean
FR185	no_closureInvariant_predicate_of_orbit_gap Paper4dFrontier/ObstructionPredicateCandidates.lean	FR186	no_admissibleNormalizationPredicate_decides_ dominantPair Paper4dFrontier/ObstructionPredicateCandidates.lean
FR187	no_admissibleNormalizationPredicate_decides_ marginBounded Paper4dFrontier/ObstructionPredicateCandidates.lean	FR188	no_admissibleNormalizationPredicate_decides_ ghostActionTwoPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean
FR189	no_admissibleNormalizationPredicate_decides_ offsetActionZeroPairCrossOne Paper4dFrontier/ObstructionPredicateCandidates.lean	FR190	admissibleCollapseLandscapeInfinity_full_paper Paper4dFrontier/ObstructionPredicateCandidates.lean
FR191	ClosureSoundPackage Paper4dFrontier/ObstructionPredicateCandidates.lean	FR192	ReasonableGuardrailPackage Paper4dFrontier/ObstructionPredicateCandidates.lean
FR193	no_closureSoundPackage_predicate_decides_four_ obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean	FR194	no_reasonableGuardrailPackage_predicate_ decides_four_obstruction_families Paper4dFrontier/ObstructionPredicateCandidates.lean
FR197	classifier_agrees_on_closureEquivalent_of_ correctOnDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR198	no_correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR199	correct_classifier_inherits_ closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean	FR200	admissibleNormalizationPredicate_has_explicit_ inhabitants Paper4dFrontier/AdmissibleCharacterization.lean
FR201	closureLawInvariant_of_iff_of_ closureEquivalent Paper4dFrontier/AdmissibleCharacterization.lean	FR202	exists_orbit_gap_of_not_closureLawInvariant Paper4dFrontier/AdmissibleCharacterization.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR203	closureLawInvariant_iff_no_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean	FR204	no_exact_closureLawInvariant_classifier_iff_ exists_orbit_gap Paper4dFrontier/AdmissibleCharacterization.lean
FR205	distinctActionCount Paper4dFrontier/DistinctActionProfiles.lean	FR206	actionCount_profileCompressedSlice Paper4dFrontier/DistinctActionProfiles.lean
FR207	decisionEquiv_profileCompressedSlice_iff Paper4dFrontier/DistinctActionProfiles.lean	FR208	profileCompressedSlice_preserves_ exactCertification Paper4dFrontier/DistinctActionProfiles.lean
FR209	profileCompressedSlice_bounded_actions Paper4dFrontier/DistinctActionProfiles.lean	FR210	OrbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR211	no_orbitGapOn_of_exact_classifiable_by_ closureLawInvariant_onDomain Paper4dFrontier/AdmissibleCharacterization.lean	FR212	exact_classifiable_by_closureLawInvariant_ onDomain_iff_no_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean
FR213	no_exact_closureLawInvariant_classifier_ onDomain_iff_orbitGapOn Paper4dFrontier/AdmissibleCharacterization.lean	FR214	boundedPatternScheme_holds_largeActionCount_ iff Paper4dFrontier/AdmissibleCharacterization.lean
FR215	boundedPatternDefinable_eventually_constant_ in_actionCount Paper4dFrontier/AdmissibleCharacterization.lean	FR216	boundedPatternDefinable_largeActionCount_ agrees Paper4dFrontier/AdmissibleCharacterization.lean
FR217	payloadSufficient_iff_realizingProblem_ isSufficient Paper4dFrontier/Realizability.lean	FR218	payloadRelevant_iff_realizingProblem_ isRelevant Paper4dFrontier/Realizability.lean
FR219	payloadIrrelevant_iff_realizingProblem_ isIrrelevant Paper4dFrontier/Realizability.lean	FR220	payloadMinimalSufficient_iff_realizingProblem_ isMinimalSufficient Paper4dFrontier/Realizability.lean
FR221	payloadSufficientSets_eq_realizingProblem_ sufficientSets Paper4dFrontier/Realizability.lean	FR222	payloadRelevantSet_eq_realizingProblem_ relevantSet Paper4dFrontier/Realizability.lean
FR223	feasiblePayloadSufficient_iff_ lawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR224	feasiblePayloadRelevant_iff_ lawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR225	feasiblePayloadIrrelevant_iff_ lawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR226	setValuedPayloadSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR227	setValuedPayloadRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR228	setValuedPayloadIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR229	outputSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR230	outputSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR231	outputSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR232	approximationSemanticsSufficient_iff_ totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR233	approximationSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR234	approximationSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR235	outputSemanticsSufficientSets_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR236	outputSemanticsRelevantSet_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean
FR237	quotientMap_realizes_setoid_asOutputSemantics Paper4dFrontier/Realizability.lean	FR240	exists_outputSemantics_realizing_setoid Paper4dFrontier/Realizability.lean
FR241	outputSemanticsSetoid Paper4dFrontier/Realizability.lean	FR242	SemanticallyExtensionalMap Paper4dFrontier/Realizability.lean
FR243	semanticallyExtensionalMap_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean	FR244	semanticallyExtensionalClaim_factors_through_ outputSemanticsQuotient Paper4dFrontier/Realizability.lean
FR245	optimizerComputation_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean	FR246	optimizerSetPayload_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean
FR247	optimizerSetSearch_polytime_ closureLawInvariant Paper4dFrontier/ComputeCostApplications.lean	FR248	optimizerComputation_polytime_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR249	optimizerComputation_polytime_classifier_agrees_on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean	FR250	no_correct_optimizerComputation_polytime_classifier_decides_dominantPair Paper4dFrontier/ComputeCostApplications.lean
FR251	no_admissibleNormalizationPredicate_optimizerComputation_polytime_and_dominantPair Paper4dFrontier/ComputeCostApplications.lean	FR252	CorrectnessForcesOrbitAgreementOnDomain Paper4dFrontier/AdmissibleCharacterization.lean
FR253	no_orbitGapOn_of_correct_classifier_onDomain_of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean	FR254	correct_classifier_onDomain_iff_no_orbitGapOn_of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean
FR255	no_correct_classifier_onDomain_iff_orbitGapOn_of_forcedOrbitAgreement Paper4dFrontier/AdmissibleCharacterization.lean	FR256	optimizerComputation_polytime_correct_classifier_onDomain_iff_no_orbitGapOn Paper4dFrontier/ComputeCostApplications.lean
FR259	optimizerSetPayload_polytime_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean	FR260	optimizerSetPayload_polytime_classifier_agrees_on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean
FR261	optimizerSetSearch_polytime_classifier_agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostApplications.lean	FR262	optimizerSetSearch_polytime_classifier_agrees_on_closureEquivalent Paper4dFrontier/ComputeCostApplications.lean
FR263	relevant_of_uniformApprox_of_strict_gap_witness Paper4dFrontier/ApproximateAdmissibility.lean	FR264	relevance_can_flip_under_arbitrarily_small_uniform_perturbation Paper4dFrontier/ApproximateAdmissibility.lean
FR266	not_sufficient_of_uniformApprox_of_strict_gap_witness Paper4dFrontier/ApproximateAdmissibility.lean	FR267	sufficiency_can_flip_under_arbitrarily_small_uniform_perturbation Paper4dFrontier/ApproximateAdmissibility.lean
FR268	pacGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR269	pacGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR270	pacGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR271	regretGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR272	regretGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR273	regretGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR274	statisticalRiskSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR275	statisticalRiskSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR276	statisticalRiskSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR277	anytimeGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean
FR278	anytimeGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR279	anytimeGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR280	finiteHorizonGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean	FR281	finiteHorizonGuaranteeSemanticsRelevant_iff_totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean
FR282	finiteHorizonGuaranteeSemanticsIrrelevant_iff_totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean	FR283	statisticalGuaranteeSemanticsSufficientSets_eq_of_outputSemanticsEquivalent Paper4dFrontier/Realizability.lean
FR284	statisticalGuaranteeSemanticsRelevantSet_eq_of_outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR285	statisticalGuarantee_classifier_agrees_on_closureEquivalent_of_correctOnDomain_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR286	no_correctOnDomain_statisticalGuarantee_classifier_of_orbit_gap_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR287	statisticalGuarantee_correct_classifier_onDomain_iff_no_orbitGapOn_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR288	opt_eq_of_uniformApprox_of_uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR289	sufficientSets_eq_of_uniformApprox_of_uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean
FR290	relevantSet_eq_of_uniformApprox_of_uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR291	randomizedGuaranteeSemanticsSufficient_iff_totalizedLawDecisionProblem_isSufficient Paper4dFrontier/Realizability.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
FR292	randomizedGuaranteeSemanticsRelevant_iff_ totalizedLawDecisionProblem_isRelevant Paper4dFrontier/Realizability.lean	FR293	randomizedGuaranteeSemanticsIrrelevant_iff_ totalizedLawDecisionProblem_isIrrelevant Paper4dFrontier/Realizability.lean
FR294	randomizedGuarantee_classifier_agrees_on_ closureEquivalent_of_correctOnDomain_of_ transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR295	no_correctOnDomain_randomizedGuarantee_ classifier_of_orbit_gap_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean
FR296	randomizedGuarantee_correct_classifier_ onDomain_iff_no_orbitGapOn_of_transfer Paper4dFrontier/StatisticalSemanticsApplications.lean	FR297	transferredSemantics_classifier_agrees_on_ closureEquivalent_of_correctOnDomain Paper4dFrontier/StatisticalSemanticsApplications.lean
FR298	no_correctOnDomain_transferredSemantics_ classifier_of_orbit_gap Paper4dFrontier/StatisticalSemanticsApplications.lean	FR299	transferredSemantics_correct_classifier_ onDomain_iff_no_orbitGapOn Paper4dFrontier/StatisticalSemanticsApplications.lean
FR300	decisionEquiv_iff_of_uniformApprox_of_ uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean	FR301	isMinimalSufficient_iff_of_uniformApprox_of_ uniformStrictGapCover Paper4dFrontier/ApproximateAdmissibility.lean
FR302	outputSemanticsExactRelevanceProfile_eq_of_ outputSemanticsEquivalent Paper4dFrontier/Realizability.lean	FR303	outputSemanticsExactRelevanceProfile_eq_ totalizedLawDecisionProblem Paper4dFrontier/Realizability.lean
FR304	agreeOn_univ_iff_eq_of_ finiteIndicatorCoordinateSpace Paper4dFrontier/Realizability.lean	FR305	finite_outputSemantics_ allCoordinatesSufficient Paper4dFrontier/Realizability.lean
FR306	finite_outputSemantics_realized_by_ exactCertification Paper4dFrontier/Realizability.lean	FR307	agreeOn_univ_iff_eq_of_ singletonIdentityCoordinateSpace Paper4dFrontier/Realizability.lean
FR308	outputSemantics_admits_coordinatePresentation Paper4dFrontier/Realizability.lean	FR311	encodable_stateSpace_admits_ countableBooleanPresentation Paper4dFrontier/Realizability.lean
FR312	encodable_discreteMeasurable_stateSpace_ admits_measurable_countableBooleanPresentation Paper4dFrontier/Realizability.lean	FR313	encodable_discreteTopological_stateSpace_ admits_continuous_countableBooleanPresentation Paper4dFrontier/Realizability.lean
FR314	agreeOn_univ_iff_eq_of_ finiteBinaryCoordinateSpace Paper4dFrontier/Realizability.lean	FR315	finite_exactSpecification_admits_ lowDimBooleanPresentation Paper4dFrontier/Realizability.lean
FR316	IdentityOutputClosureSpec.classifier_agrees_ on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR317	IdentityOutputClosureSpec.no_correctOnDomain_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR318	hypothesisOutput_classifier_agrees_on_ closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR319	estimatorOutput_classifier_agrees_on_ closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR320	policyOutput_classifier_agrees_on_ closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR321	randomizedProcedure_classifier_agrees_on_ closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR322	no_correctOnDomain_hypothesisOutput_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR323	no_correctOnDomain_estimatorOutput_classifier_ of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR324	no_correctOnDomain_policyOutput_classifier_of_ orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR325	no_correctOnDomain_randomizedProcedure_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR326	TransportedOutputClosureSpec.classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR327	TransportedOutputClosureSpec.no_ correctOnDomain_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR328	IdentityOutputClosureSpec.correctOnDomain_iff_ correctOnDomain_transport Paper4dFrontier/ComputeCostExternalOutputs.lean	FR329	representationRelativeHypothesis_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR330	representationRelativeEstimator_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR331	representationRelativePolicy_classifier_ agrees_on_closureEquivalent_of_correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean
FR332	representationRelativeRandomizedProcedure_ classifier_agrees_on_closureEquivalent_of_ correctOnDomain Paper4dFrontier/ComputeCostExternalOutputs.lean	FR333	no_correctOnDomain_representationRelativeHypothesis_ classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean

(continued...)

ID	Lean Handle / Source	ID	Lean Handle / Source
FR334	no_correctOnDomain_representationRelativeEstimator_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR335	no_correctOnDomain_representationRelativePolicy_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean
FR336	no_correctOnDomain_representationRelativeRandomizedProcedure_classifier_of_orbit_gap Paper4dFrontier/ComputeCostExternalOutputs.lean	FR340	booleanPayloadTransfer Paper4dFrontier/Realizability.lean
FR341	predicateTransfer Paper4dFrontier/Realizability.lean	FR342	sufficiency_is_relation_refinement Paper4dFrontier/Realizability.lean
FR343	relevance_is_erased_failure_of_refinement Paper4dFrontier/Realizability.lean	FR345	exactSemanticsQuotient_universal_characterization Paper4dFrontier/Realizability.lean
FR346	exactnessMeansExactAgreementWithValidity Paper4dFrontier/Realizability.lean		

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary 6.44: Exact Approximation Semantics Transfer	FR234, FR233, FR232
Corollary 6.26: Boolean Payload Transfer	FR340
Corollary 6.6: Constant-Optimizer Collapse Forces Trivial Certification	FR52, FR53
Corollary 4.14: Domain Restriction Helps Only by Removing Orbit Gaps	FR252, FR254, FR255, FR253, FR256
Corollary 6.50: Every State Equivalence Relation Is Realizable as Exact Output Semantics	FR240, FR237
Corollary 6.20: Compute-Cost Version: External Output Objects	FR316, FR328, FR317, FR326, FR327, FR319, FR318, FR323, FR322, FR324, FR325, FR320, FR321
Corollary 2.3: Finite Basis for the Current Positive Landscape	FR3, FR29
Corollary 4.11: No Correct Tractability Classifier	FR197, FR199, FR193, FR198
Corollary 4.13: Orbit-Gap Completeness on Closure-Closed Domains	FR210, FR212, FR213, FR211
Corollary 6.19: Compute-Cost Version: Canonical Payload and Search Tasks	FR260, FR259, FR262, FR261
Corollary 6.27: Predicate Transfer	FR341
Corollary 6.42: Randomized-Output Guarantee Semantics Inherit the Closure-Orbit Consequences	FR295, FR293, FR292, FR291, FR294, FR296
Corollary 6.41: Randomized-Output Guarantee Semantics Transfer	FR293, FR292, FR291
Corollary 5.4: Realizability Alone Cannot Be the Frontier	FR7, FR8
Corollary 4.10: Reasonable Guardrail Packages	FR194
Corollary 6.30: Arbitrary Exact Output Semantics Transfer	FR231, FR230, FR229
Corollary 6.3: Relevance Is Failure of Erased Sufficiency	FR49, FR50
Corollary 6.33: Relevance Is Erased Failure of Refinement	FR343
Corollary 6.28: Nonempty Set-Valued Payload Transfer	FR225, FR224, FR223
Corollary 6.46: Arbitrarily Small Uniform Perturbations Can Flip Relevance	FR264
Corollary 6.47: Arbitrarily Small Uniform Perturbations Can Flip Sufficiency	FR267

Paper claim	**Lean handle**
Corollary 6.43: Statistical Guarantee Semantics Inherit the Closure-Orbit Consequences	FR279, FR278, FR277, FR282, FR281, FR280, FR295, FR286, FR298, FR288, FR270, FR269, FR268, FR293, FR292, FR291, FR294, FR296, FR273, FR272, FR271, FR290, FR284, FR283, FR285, FR287, FR276, FR275, FR274, FR289, FR297, FR299
Corollary 6.40: Statistical Guarantee Semantics Transfer	FR279, FR278, FR277, FR282, FR281, FR280, FR270, FR269, FR268, FR273, FR272, FR271, FR276, FR275, FR274
Corollary 6.4: Certification Statistics Factor Through the Quotient Relation	FR41, FR43, FR44, FR42
Corollary 6.2: Sufficiency Is Relation Refinement	FR48
Corollary 6.29: Arbitrary Set-Valued Payload Transfer via Failure Tokens	FR228, FR227, FR226
Corollary 6.21: Compute-Cost Version: Representation-Relative Output Objects	FR328, FR326, FR327, FR334, FR333, FR335, FR336, FR330, FR329, FR331, FR332
Proposition 6.22: The Admissibility Layer Is Nonempty	FR200
Proposition 6.8: Positive Affine Utility Reparameterizations Are Invisible	FR13, FR16, FR15, FR14
Proposition 6.45: Approximate Relevance and Sufficiency Claims Need Explicit Stability Control	FR266, FR263
Proposition 4.1: Binary Pairwise Symmetry Dichotomy	FR120, FR118, FR119
Proposition 6.24: Bounded-Pattern Predicates Stabilize Above a Finite Action Bound	FR215, FR216, FR214
Proposition 6.35: Canonical Exact Relevance Profile	FR302, FR303
Proposition 6.15: Closure Operations Preserve Exact Certification	FR57, FR17, FR18, FR60, FR56, FR59, FR84, FR16, FR55, FR58, FR83, FR15, FR19, FR20, FR85
Proposition 6.38: Countable Exact Specifications Admit Countable Boolean Presentations	FR312, FR313, FR311
Proposition 4.2: Decision-Relevant Binary Pairwise Dichotomy	FR121, FR123, FR124, FR122
Proposition 3.2: Degenerate Positive Cases	FR1, FR6
Proposition 6.1: Exact Certification Depends Only on the Decision Quotient Relation	FR32, FR33, FR31, FR30, FR37, FR51, FR35, FR34
Proposition 6.23: Bounded Distinct Action Profiles Compress to Bounded Actions	FR206, FR207, FR205, FR209, FR208
Proposition 6.9: Duplicate Actions and Duplicate States Preserve Certification	FR71, FR68, FR54, FR70, FR56, FR69, FR55, FR73, FR72
Proposition 6.7: Explicit Enumeration Is Parameter-Dependent, Not Structural	FR66, FR67
Proposition 6.36: Every Exact Specification Admits a Coordinate Presentation	FR307, FR308
Proposition 6.31: Exactness Means Exact Agreement With Validity	FR346
Proposition 2.4: Explicit Primitive Basis	FR29
Proposition 2.2: Explicit Inventory	FR26, FR25, FR3, FR28, FR27
Proposition 6.37: Finite Exact Specifications Admit Coordinate Presentations	FR304, FR305, FR306
Proposition 6.39: Finite Exact Specifications Admit Low-Dimensional Boolean Presentations	FR314, FR315

Paper claim	**Lean handle**
Proposition 2.5: All Independent Core Mechanisms Are Nondegenerate	FR108, FR106, FR112, FR101, FR105, FR113
Proposition 6.48: Global Approximation Stability Under Uniform Strict Gaps	FR300, FR301, FR295, FR298, FR288, FR293, FR292, FR291, FR294, FR296, FR290, FR289, FR297, FR299
Proposition 2.9: Current Positive Interfaces Factor Through Role Classification	FR4
Proposition 6.12: Invariance Alone Is Too Weak	FR22, FR21, FR23
Proposition 2.6: Weak Tree Ordering Does Not Imply Width One	FR115, FR114
Proposition 3.1: Lifted Cases Introduce No New Primitive	FR5
Proposition 6.14: Unrestricted Predicates Trivialize Exact Characterization	FR145
Proposition 4.12: Orbit-Gap Completeness for Exact Classification	FR203, FR201, FR202, FR204
Proposition 4.3: Orbit-Gap Template	FR185
Proposition 2.8: Parent-Tree Structure Gives Width-One Decompositions	FR110, FR100, FR111
Proposition 6.25: Deterministic Payload Transfer	FR219, FR220, FR222, FR218, FR221, FR217
Proposition 6.11: Quotient Size Is Unbounded Under Realizability	FR24, FR38
Proposition 5.3: The Realized Quotient Is Exactly the Label Range	FR39, FR40
Proposition 6.10: Relabeling Invariance Is Forced by Exact Certification	FR17, FR18, FR19, FR20
Proposition 6.49: Exact Semantic Claims Depend Only on Admissible-Output Equivalence	FR236, FR235
Proposition 6.51: Semantically Extensional Claims Factor Through the Exact-Semantics Quotient	FR242, FR241, FR244, FR243
Proposition 5.2: Every Equivalence Relation Is Optimizer-Realizable	FR47, FR46, FR45
Proposition 6.13: Independent Summary Frameworks Converge on the Same Structural Core	FR178, FR181, FR182, FR180, FR179
Proposition 6.32: Sufficiency Is Relation Refinement	FR342
Proposition 2.7: Cyclic Dependencies Recover Hardness	FR110, FR100, FR111, FR115
Proposition 6.5: Zero-Distortion Summaries Refine the Optimizer Quotient	FR177, FR176
Theorem 4.8: Admissible Collapse Across the Four Obstruction Families	FR190
Theorem 4.9: Closure-Sound Package No-Go	FR193
Definition 6.17: Admissible Normalization Predicate	FR197, FR199, FR198, FR249, FR248, FR245, FR246, FR247
Theorem 2.1: Explicit Partition of the Current Positive Landscape	FR26, FR25, FR1, FR2, FR28, FR27
Theorem 4.4: Dominant-Pair Admissible No-Go	FR186
Theorem 6.34: Exact Semantics Quotient Universality	FR345
Theorem 4.6: Ghost-Action Admissible No-Go	FR188
Theorem 5.1: Every Labeling Kernel Is Optimizer-Realizable	FR7, FR8
Theorem 4.5: Margin-Masking Admissible No-Go	FR187
Theorem 4.7: Offset Admissible No-Go	FR189
Theorem 6.18: Compute-Cost Version: Optimizer Computation	FR248

Auto summary: mapped 79/79 (full=79, derived=0, unmapped=0).